Exp 11.1

→ Write a program to apply bitwise OR, AND and NOT operators on bit level

→

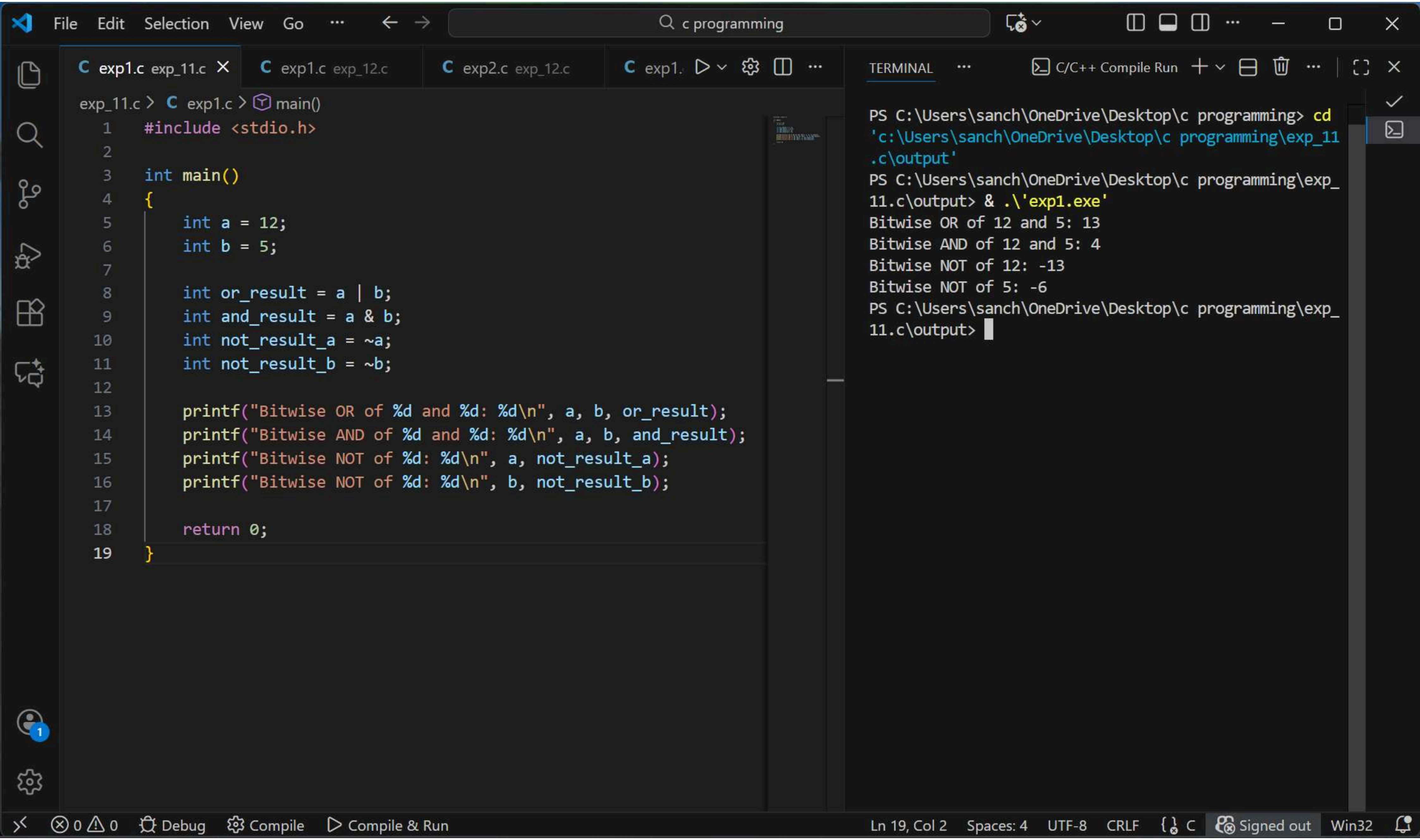
```
#include <stdio.h>

int main ()
{
  int a = 12 ;
  int b = 5 ;

int or_result = a|b ;
int and_result = a & b ;
int not_result_a = ~a ;
int not_result_b = ~b ;

printf ("Bitwise OR of %d and %d : %d\n", a, b,
 or_result) ;
printf ("Bitwise AND of %d : %d\n", a, b,
 and_result ) ;
printf ("Bitwise NOT of %d : %d\n", a, not_result_a);
printf ("Bitwise NOT of %d : %d\n, b, not_result_b);

return 0;
}
```


File Edit Selection View Go
c programming
exp1.c exp_11.c
exp1.c exp_12.c
exp2.c exp_12.c
exp_11.c > exp1.c > main()
#include <stdio.h>
int main()
{
int a = 12;
int b = 5;
int or_result = a | b;
int and_result = a & b;
int not_result_a = ~a;
int not_result_b = ~b;
printf("Bitwise OR of %d and %d: %d\n", a, b, or_result);
printf("Bitwise AND of %d and %d: %d\n", a, b, and_result);
printf("Bitwise NOT of %d: %d\n", a, not_result_a);
printf("Bitwise NOT of %d: %d\n", b, not_result_b);
return 0;
}
TERMINAL
C/C++ Compile Run
PS C:\Users\sanch\OneDrive\Desktop\c programming> cd 'c:\Users\sanch\OneDrive\Desktop\c programming\exp_11.c\output'
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_11.c\output> & .\'exp1.exe'
Bitwise OR of 12 and 5: 13
Bitwise AND of 12 and 5: 4
Bitwise NOT of 12: -13
Bitwise NOT of 5: -6
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_11.c\output>
Debug Compile Compile & Run
Ln 19, Col 2 Spaces: 4 UTF-8 CRLF C Signed out Win32

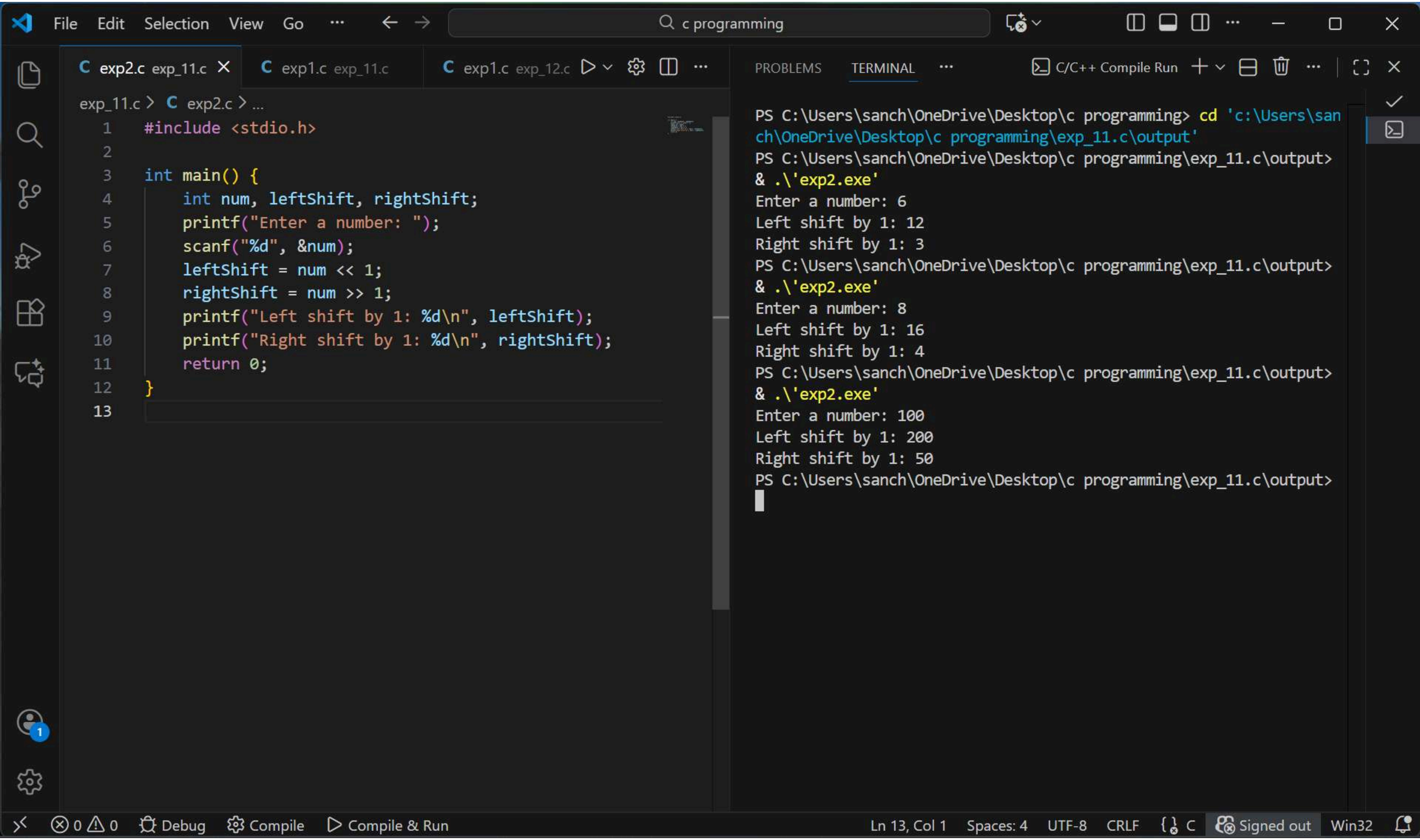
Exp 11.2

→ Write a program to apply left shift and right shift operator.

→

```
#include <stdio.h>

int main () {
 int num, leftshift, rightshift;
  printf ("Enter a number: ");
  scanf ("%d", &num);
  leftshift = num << 1;
  rightshift = num >> 1;
  printf ("Left shift by 1: %d\n", leftshift);
  printf ("Right shift by 1: %d\n", rightshift);
  return 0;
}
```


File Edit Selection View Go
c programming
exp2.c exp_11.c
exp1.c exp_11.c
exp1.c exp_12.c
PROBLEMS
TERMINAL
C/C++ Compile Run
exp_11.c > C exp2.c > ...
#include <stdio.h>
int main() {
int num, leftShift, rightShift;
printf("Enter a number: ");
scanf("%d", &num);
leftShift = num << 1;
rightShift = num >> 1;
printf("Left shift by 1: %d\n", leftShift);
printf("Right shift by 1: %d\n", rightShift);
return 0;
}
PS C:\Users\sanch\OneDrive\Desktop\c programming> cd 'c:\Users\sanch\OneDrive\Desktop\c programming\exp_11.c\output'
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_11.c\output> & .\'exp2.exe'
Enter a number: 6
Left shift by 1: 12
Right shift by 1: 3
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_11.c\output> & .\'exp2.exe'
Enter a number: 8
Left shift by 1: 16
Right shift by 1: 4
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_11.c\output> & .\'exp2.exe'
Enter a number: 100
Left shift by 1: 200
Right shift by 1: 50
PS C:\Users\sanch\OneDrive\Desktop\c programming\exp_11.c\output>
0 0 Debug Compile Compile & Run
Ln 13, Col 1 Spaces: 4 UTF-8 CRLF C Signed out Win32